

Hossam Abu-elez Fathy

Machine Learning Engineer | Data Analyst | Data Scientist
Bachelor of Computer Science & Artificial Intelligence

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🐙 GitHub

🌐 LinkedIn

🌐 Portfolio

PROFILE

Computer Science and Artificial Intelligence student with hands-on experience in Machine Learning, Data Analysis, and Data Science. Experienced in building end-to-end AI solutions using Python, Scikit-Learn, SQL, and data visualization tools. Developed machine learning models achieving up to 99.46% accuracy on real-world datasets. Seeking internships and entry-level opportunities in Machine Learning, Data Analytics, and Data Science.

EDUCATION

•Artificial Intelligence Sciences

New Mansoura University

Oct 2023 – Jun 2027

CGPA: 3.4 / 4.0

EXPERIENCE

•Machine Learning Engineering Intern

FlyRank AI

Jul 2026 – Present

Remote

- Developing machine learning solutions for real-world AI applications.
- Building, evaluating, and improving ML models using Python.
- Collaborating with engineering teams on AI-driven projects.

•Data Analytics & BI Trainee

Information Technology Institute (ITI)

Jul 2026 – Present

Mansoura

- Building dashboards using Power BI and Tableau.
- Writing SQL queries and performing data analysis using Excel.
- Learning Data Warehousing and Business Intelligence concepts.

•Machine Learning Trainee

Creativa Innovation Hub

Jun 2025 – Aug 2025

Mansoura

- Preprocessed datasets and built ML models using Scikit-Learn.
- Applied feature engineering, classification, and regression techniques.
- Visualized data insights using Matplotlib and Plotly.

TECHNICAL SKILLS

Programming: Python, C++

Machine Learning: Supervised Learning, Unsupervised Learning, Deep Learning, Neural Networks, Machine Learning Algorithms, Model Evaluation, Model Selection, Feature Engineering, Data Preprocessing

Libraries: Scikit-Learn, Pandas, NumPy, Matplotlib, Plotly

Data Analysis: Excel, Power BI, EDA, Data Cleaning, Data Visualization

Databases: SQL, MySQL, PostgreSQL

Software Engineering: REST APIs, Git, GitHub, Object-Oriented Programming, Software Engineering Methodologies, Full-Stack Development

Tools: Git, GitHub, Markdown, Jupyter Notebook, VS Code, Kaggle

Soft Skills: Problem Solving, Communication, Team Collaboration, Analytical Thinking, Presentation, Adaptability, Continuous Learning

PROJECTS

•Analytics of Jumia Product Reviews

2026

Data Analytics & Business Intelligence Project

- Analyzed **3,809** Jumia product reviews covering 877 unique products using SQL Server, Python, and Power BI.
- Performed data exploration, validation, cleaning, and text preprocessing using SQL Server, Pandas, and NumPy, including Arabic and English review analysis.
- Built an interactive Power BI dashboard with data modeling, KPIs, filters, drill-through, product performance, and customer feedback analysis.
- **Technologies:** Python, Pandas, NumPy, SQL Server, Power BI, Excel, Text Analytics.

•MedOptima AI Platform

2025

AI Healthcare Decision Support System

- Developed an end-to-end AI healthcare decision support platform using Neural Networks, Knowledge-Based Systems, and Optimization Algorithms.
- Achieved **99.46%** test accuracy on a Kaggle dataset containing 41 diseases and 117 symptoms.
- Performed data preprocessing, feature engineering, hyperparameter tuning, and model evaluation.
- **Technologies:** Python, Scikit-Learn, SQL, REST APIs, Neural Networks, Full-Stack Development.

•AI Optimization Research Project - FTT Algorithm

2025

Research and Optimization

- Designed and implemented the Football Team Training (FTT) Optimization Algorithm.
- Solved supply chain optimization problems and compared the proposed algorithm with existing methods.
- Improved convergence speed and solution quality.

•Air Quality Prediction using Machine Learning

2024 – 2025

Machine Learning Regression Project

- Collected, cleaned, and preprocessed environmental datasets for predictive modeling.
- Built and evaluated regression models using Python and Scikit-Learn.
- Performed EDA and created interactive visualizations using Plotly and Matplotlib.

•ALU 4-bit Digital Design Project

2024

Computer Architecture Project

- Designed and implemented a 4-bit ALU supporting arithmetic and logical operations.
- Simulated and tested the ALU using digital logic design tools.

•Computer Architecture & Assembly - 8085 & 8086

2024

Assembly Programming

- Implemented and analyzed assembly programs using 8085 and 8086 instruction sets.
- Analyzed processor architecture, memory organization, and instruction execution.

•Airline Reservation System

2023 – 2024

Full Software Engineering Project

- Designed system requirements, UML diagrams, and system architecture.
- Applied Object-Oriented Programming (OOP) principles and software engineering methodologies.
- Integrated a relational database for efficient data management.

CERTIFICATES

•Freelancing Basics

Sep 2026

•Find Your Freelance Fit

Sep 2026

•Getting Started as a Freelancer (Upwork)

Sep 2026

•Getting Started as a Freelancer (Khamsat)

Sep 2026

•Getting Started as a Freelancer (Mostaql)

Sep 2026

•Machine Learning Training

Jun 2025

•Transact SQL Queries Using SQL Server

Mar 2025

•Communicate Effectively on GitHub using Markdown

Apr 2024

LEADERSHIP & VOLUNTEERING

- **AI & Programming Student Mentor** Supporting Students and Beginners *Present*
 - Mentored students in AI, Python, and programming fundamentals.
 - Guided beginners in problem-solving and technical skill development.
- **Tech Event Organizer** AI & Technology Workshops *Present*
 - Organized AI and technology workshops for students.
 - Coordinated event logistics and supported successful workshop execution.

LANGUAGES

- **Arabic:** Native
- **English:** Proficient